

Prepare in priority as mentioned below:

- Programming + OOP
- DSA(Easy)
- SQL + DBMS
- SDLC + Agile
- Git
- Cloud Fundamentals
- React/Angular
- Basic Design Principles

Basic Coding Problems

1. Reverse a String
2. Check for Palindrome
3. Find the Maximum Element in an Array
4. Count the Number of Vowels in a String
5. Find the Sum of Digits of a Number
6. Find the Nth Fibonacci Number
7. Print All Prime Numbers up to N
8. Remove Duplicates from a String
9. Find the Second Largest Element in an Array
10. Check if Two Strings Are Anagrams
11. Check Whether a Year Is a Leap Year
12. Swap Two Numbers Without Using a Third Variable
13. Check Whether a Number Is an Armstrong Number
14. Convert Decimal to Binary and Binary to Decimal
15. Find GCD and LCM of Two Numbers
16. Find the Frequency of Each Number in an Array
17. Linear Search
18. Binary Search
19. Find the Minimum Element in an Array
20. Reverse an Integer

21. Check Whether a Number Is Prime
 22. Find Factorial of a Number
 23. Count the Number of Digits in a Number
 24. Find the Sum of Elements in an Array
 25. Find Duplicate Elements in an Array
 26. Find the Missing Number in an Array
 27. Count Even and Odd Numbers in an Array
 28. Find the Largest and Smallest Element in an Array
 29. Check Whether a String Contains Only Digits
 30. Count Frequency of Each Character in a String
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Topic-Wise DSA Problems

Arrays & Strings

1. Contains Duplicate
<https://leetcode.com/problems/contains-duplicate/description/>
2. Kadane's Algorithm
<https://www.geeksforgeeks.org/problems/kadanes-algorithm-1587115620/1>
3. Best Time to Buy and Sell Stock
<https://leetcode.com/problems/best-time-to-buy-and-sell-stock/description/>
4. Count Pairs with Given Sum
<https://www.geeksforgeeks.org/problems/count-pairs-with-given-sum5022/1>
5. Subarray with 0 Sum
<https://www.geeksforgeeks.org/problems/subarray-with-0-sum-1587115621/1>
6. Two Sum
<https://leetcode.com/problems/two-sum/description/>
7. Triplet Sum in Array
<https://www.geeksforgeeks.org/problems/triplet-sum-in-array-1587115621/1>
8. Move All Negative Numbers to One Side
<https://www.geeksforgeeks.org/problems/move-all-negative-elements-to-end1813/1>
9. Valid Anagram
<https://leetcode.com/problems/valid-anagram/description/>
10. Reverse Words in a String
<https://www.geeksforgeeks.org/problems/reverse-words-in-a-given-string5459/1>

11. Replace Element with Greatest Element on Right Side

<https://leetcode.com/problems/replace-elements-with-greatest-element-on-right-side/description/>

12. Find Missing and Repeated Values

<https://leetcode.com/problems/find-missing-and-repeated-values/description/>

13. Reverse String

<https://leetcode.com/problems/reverse-string/description/>

14. Valid Palindrome

<https://leetcode.com/problems/valid-palindrome/description/>

Binary Search

15. Binary Search

<https://leetcode.com/problems/binary-search/description/>

Priority Queue

16. Kth Largest Element

<https://leetcode.com/problems/kth-largest-element-in-an-array/description/>

Linked List

17. Linked List Operations

<https://www.programiz.com/dsa/linked-list-operations>

18. Reverse Linked List

<https://leetcode.com/problems/reverse-linked-list/description/>

19. Linked List Cycle

<https://leetcode.com/problems/linked-list-cycle/description/>

20. Palindrome Linked List

<https://leetcode.com/problems/palindrome-linked-list/description/>

21. Remove Duplicates from Sorted Linked List

<https://leetcode.com/problems/remove-duplicates-from-sorted-list/description/>

22. Middle of Linked List

<https://leetcode.com/problems/middle-of-the-linked-list/description/>

23. Delete Nodes Having Greater Value on Right

<https://www.geeksforgeeks.org/problems/delete-nodes-having-greater-value-on-right/1>

Matrix

24.Transpose Matrix

<https://leetcode.com/problems/transpose-matrix/description/>

25.Spiral Traversal of Matrix

<https://www.geeksforgeeks.org/problems/spirally-traversing-a-matrix-1587115621/1>

Data Structures

1. What is linear and non-linear data structure?
2. What is a stack data structure? What are the operations available on stack?
3. What is a Queue data structure? What are the operations available on queue?
4. What is array data structure?
5. What is a linked list and its types?
6. What are the advantages of a linked list over an array?
7. In which scenarios do we use Linked List and when Array?
8. What is an asymptotic analysis of an algorithm?
9. What is a hashmap in data structure and its time complexity?
10. What is binary tree data structure?
11. What is binary search tree data structure?
12. What are tree traversals? Inorder, preorder, postorder.
13. What is a priority queue?
14. What is graph data structure and its representations?
15. What is the difference between the Breadth First Search (BFS) and Depth First Search (DFS)?
16. What is AVL tree data structure?
17. Define Red-Black Tree.
18. What is a heap data structure?
19. Which sorting algorithm is considered the fastest? Why?
20. Learn best case, average case, worst case time complexity of Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort.
21. What is Merge Sort? How does it work?
22. What are the advantages of a binary search over a linear search?
23. What is the time complexity of binary search?
24. What is hashing? What are the different techniques of hashing?
25. What is the time complexity of search, insert, and delete operations in a Hash Table?
26. What is recursion?

27. What is the time complexity to search in an unsorted and sorted array?
28. What is the time complexity of Linked List operations? (insertion, deletion, search, traversal)
29. What approach would you take to find the shortest route between two locations? (Dijkstra algorithm)

DBMS Interview Questions

1. What is the difference between WHERE and HAVING?
2. Write a query to fetch the second-highest salary from a table using:
 - o Subquery
 - o LIMIT / OFFSET
 - o RANK()
3. What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and OUTER JOIN?
4. Find the number of employees in each department.
5. What are keys in DBMS?
 - o Primary Key
 - o Candidate Key
 - o Alternate Key
 - o Foreign Key
 - o Composite Key
 - o Super Key
6. What are PRIMARY KEY and UNIQUE constraints? What is the difference between them?
7. What is referential integrity?
8. What is a VIEW in SQL?
9. Write a query to fetch the top 3 highest-paid employees using:
 - o LIMIT / OFFSET
 - o RANK()
10. What is the difference between DELETE, TRUNCATE, and DROP?
11. What is an Entity-Relationship Diagram (ERD)?
12. What is normalization? Why is it important? Explain the various normal forms.
13. Explain the different types of relationships among tables in a DBMS.
14. What are tuples and attributes in DBMS?
15. Find employees who earn more than the average salary.
16. What is the difference between UNION and UNION ALL?

17. What is the DISTINCT keyword in SQL?
18. What is the purpose of the GROUP BY clause?
19. What is the difference between IN, EXISTS, and NOT EXISTS?
20. What are indexes? What is the difference between clustered and non-clustered indexes?
21. How does an index improve query performance? What are the disadvantages of indexes?
22. What is a transaction? What are ACID properties in a transaction?
23. What is a B-tree/B+ tree index?
24. What is pattern matching in SQL? Explain SQL wildcards.
25. What is the difference between OLTP and OLAP?
26. What is DDL? Explain:
 - CREATE
 - ALTER
 - DROP
 - TRUNCATE
23. What is DML? Explain:
 - SELECT
 - INSERT
 - UPDATE
 - DELETE
24. What is DCL? Explain:
 - GRANT
 - REVOKE
25. What is TCL? Explain:
 - COMMIT
 - ROLLBACK
 - SAVEPOINT

SQL Query-Based Questions

Basic SELECT & Filtering

1. Retrieve all columns from the Employees table.
2. Get names and salaries of employees from the IT department.
3. Show employees with salary greater than 60,000.

4. List employees who joined after 2022-01-01.
5. Get all distinct department IDs from Employees.

Sorting & Limiting

6. Show employees sorted by salary in descending order.
7. Get the top 3 highest-paid employees.

Aggregate Functions

8. Count the total number of employees.
9. Find the average salary in each department.
10. Find the maximum salary in the company.

SQL Query-Based Questions

11. Fetch the second-highest salary using:
 - Subquery
 - LIMIT/OFFSET
 - RANK() / DENSE_RANK()
12. Demonstrate INNER JOIN.
13. Demonstrate LEFT JOIN.
14. Demonstrate RIGHT JOIN.
15. Demonstrate FULL OUTER JOIN.
16. Find the number of employees in each department.
17. Fetch the top 3 highest-paid employees using LIMIT/OFFSET or RANK().
18. Find employees who earn more than the average salary.

Joins

19. Show each employee's name along with their department name.
20. List employees along with their manager's name.
21. Perform a LEFT JOIN operation for the given two tables.

Subqueries

22. Find employees earning more than the average salary.
23. Get the department name of the highest-paid employee.

String & Date Functions

24. Show employee names in uppercase.
25. Get the year of joining for each employee.

Orders Table Queries

26. Show all orders placed in August 2024.
27. Find the total sales amount.

28. Get the customer who spent the most.
29. List customers who placed more than one order.

SQL Window Functions & Query Practice

RANK(), DENSE_RANK(), ROW_NUMBER()

30. Assign a unique row number to each employee based on salary in descending order using ROW_NUMBER().
31. Assign ranks to employees based on salary using RANK(). Observe what happens when two employees have the same salary.
32. Assign dense ranks to employees based on salary using DENSE_RANK(). Compare the output with RANK().
33. Display RANK(), DENSE_RANK(), and ROW_NUMBER() for all employees and compare their outputs.

Top-N / Highest Salary

34. Retrieve the top 3 highest-paid employees using ROW_NUMBER().
35. Use DENSE_RANK() to find employees having the second-highest salary.
36. Use DENSE_RANK() to find employees having the third-highest salary.
37. Retrieve all employees having the second-highest salary.
38. Retrieve employees belonging to the top 3 salary groups.

Department-wise Window Functions

39. Find the highest-paid employee from each department.
40. Find the second-highest-paid employee from each department.
41. Find the second-lowest-salary employee in each department.
42. Display each employee along with the total salary of their department.
43. Find employees earning more than their department's average salary.
44. Find departments having more than 2 employees.

CASE / Salary Categorization

45. Write a SQL query to display each employee's name, salary, and a new column called Salary_Level:

- High → salary $\geq$ 80,000
- Medium → salary between 60,000 and 79,999
- Low → salary $<$ 60,000

Operating Systems

1. What is a bootstrap program in OS?

2. Difference between main memory and secondary memory.
 3. What is thrashing in OS?
 4. Difference between multitasking and multiprocessing OS.
 5. Difference between paging and segmentation.
 6. Difference between internal fragmentation and external fragmentation.
 7. What is virtual memory?
 8. What is a thread in OS?
 9. What is a process?
 10. What are the different states of a process?
 11. Difference between process and thread.
 12. What are the various sections of a process?
 13. What is a scheduling algorithm?
 14. Explain different CPU scheduling algorithms.
 15. Which scheduling algorithm gives the best throughput?
 16. What is process synchronization?
 17. Explain synchronization techniques.
 18. What is a critical section?
 19. What is preemptive vs. non-preemptive scheduling?
 20. What is the kernel in OS?
 21. What is a semaphore?
 22. What are the types of semaphores?
 23. Difference between semaphore and mutex.
 24. What is context switching?
 25. What is a deadlock?
 26. What are the necessary conditions for deadlock?
 27. What is starvation?
 28. Difference between starvation and deadlock.
 29. What is Belady's anomaly?
 30. Explain demand paging.
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OOPs Concepts

1. What are the main features of OOP?
2. What are access specifiers and what is their significance?
3. What are the various types of inheritance?
4. What is polymorphism?
5. What are the types of polymorphism?

6. Difference between overloading and overriding.
 7. What is data abstraction?
 8. What is an abstract class?
 9. What is a class?
 10. What is an object?
 11. What is encapsulation?
 12. What is a constructor?
 13. What are the various types of constructors?
 14. What is a copy constructor?
 15. What is a destructor?
 16. Difference between a structure and a class in C++.
 17. What are friend functions?
 18. What is a virtual function?
 19. What is a pure virtual function?
 20. What are static data members in C++?
 21. What is enumeration?
 22. What is the this pointer?
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Computer Networks

1. What are the different types of network topology?
2. What is DNS?
3. What is the use of a router?
4. Difference between a router and a gateway.
5. What is SMTP?
6. Explain the 7 layers of the OSI model.
7. What protocols are associated with each OSI layer?
8. What are HTTP and HTTPS?
9. Difference between HTTP and HTTPS.
10. What is TCP?
11. What is UDP?
12. Compare TCP and UDP.
13. What is ICMP?
14. What is ARP?
15. Difference between MAC address and IP address.
16. What is a port number?
17. What is a socket?

18. What is subnetting?
 19. Difference between a hub and a switch.
 20. What is a firewall?
 21. What are the different classes of IPv4?
 22. What are unicast, multicast, broadcast and anycast?
 23. Identify the class, network IP address, direct broadcast address and limited broadcast address for 1.2.3.4.
 24. Identify the class, network IP address, direct broadcast address and limited broadcast address for 10.15.20.6.
 25. Which protocols work at the application layer?
 26. How many hosts are available in a /29 subnet?
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Software Engineering

1. What is SDLC?
2. Explain the phases of SDLC.
3. Waterfall vs. Agile.
4. What is Agile?
5. What are the principles of Agile?
6. What is Scrum?
7. Explain Scrum roles.
8. Explain Scrum ceremonies/events.
9. Git vs. GitHub.
10. Explain the basic Git workflow: clone → branch → add → commit → push → pull request.
11. What is a merge conflict?
12. How do you resolve a merge conflict?
13. What is software testing?
14. Why is software testing important?
15. Unit testing vs. integration testing vs. system testing.
16. Regression testing vs. smoke testing vs. sanity testing.
17. What is CI/CD?
18. Continuous Integration vs. Continuous Delivery vs. Continuous Deployment.
19. What is a CI/CD pipeline?
20. Debugging vs. testing.
21. How would you debug a bug that you cannot reproduce?

Python Interview Questions

1. What are the key features of Python?
2. Is Python compiled or interpreted? Explain.
3. What are the built-in data types in Python?
4. Difference between a list and a tuple.
5. Difference between a list, tuple, set and dictionary.
6. What are mutable and immutable objects?
7. Give examples of mutable and immutable objects.
8. Difference between == and is in Python.
9. Difference between append() and extend().
10. Difference between sort() and sorted().
11. What is list comprehension?
12. Give an example of list comprehension.
13. What are *args and **kwargs?
14. What is a lambda function?
15. Difference between break, continue and pass.
16. What is self in Python?
17. What is __init__()?
18. Explain OOP concepts in Python:
 - Encapsulation
 - Inheritance
 - Polymorphism
 - Abstraction
20. How is exception handling done in Python?
21. Explain try, except, else and finally.
22. Difference between shallow copy and deep copy.
23. What are iterators?
24. What are generators?
25. What is the use of yield?
26. What is the Global Interpreter Lock (GIL) in Python?

React JS Interview Questions

1. What is React JS?

2. What are the features of React?
3. What is Virtual DOM?
4. What is the difference between Virtual DOM and Real DOM?
5. What are components in React?
6. What is the difference between functional and class components?
7. What are props in React?
8. What is state in React?
9. What is the difference between props and state?
10. What is the useState hook?
11. What is the useEffect hook?
12. What is the dependency array in useEffect?
13. What is conditional rendering?
14. What is JSX?
15. What is the difference between controlled and uncontrolled components?
16. What is lifting state up?
17. What is prop drilling?
18. What is Context API?
19. What is Redux?
20. What is the difference between Context API and Redux?
21. What is React Router?
22. What is lazy loading in React?
23. What are React Hooks?
24. What is useRef?
25. What are useMemo and useCallback?
26. What is reconciliation in React?
27. What are keys in React and why are they important?
28. What is a Higher-Order Component (HOC)?
29. What is React Fragment?
30. What is the difference between npm and npx?
31. What is hydration in React?
32. What is Server-Side Rendering (SSR)?
33. What is the difference between CSR and SSR?
34. What are React lifecycle methods?
35. What is an Error Boundary?

Cloud Services -

Understand **cloud fundamentals + one cloud platform**(AWS or Azure or GCP)

AWS Interview Questions

1. What is EC2?
2. What is Snowball?
3. What is CloudWatch?
4. What is Elastic Transcoder?
5. What do you understand by VPC?
6. DNS and Load Balancer Services come under which type of Cloud Service?
7. What are the Storage Classes available in Amazon S3?
8. Explain what T2 instances are.
9. What are Key Pairs in AWS?
10. How many Subnets can you have per VPC?
11. List different types of Cloud Services.
12. Explain what S3 is.
13. How does Amazon Route 53 provide high availability and low latency?
14. How can you send a request to Amazon S3?
15. What does AMI include?
16. What are the different types of EC2 instances?
17. What is the relation between an Availability Zone and a Region?
18. How do you monitor Amazon VPC?
19. What are the different types of EC2 instances based on their costs?
20. What do you understand by stopping and terminating an EC2 instance?
21. What are the consistency models for modern databases offered by AWS?
22. What is Geo-Targeting in CloudFront?
23. What are the advantages of AWS IAM?
24. What do you understand by a Security Group?
25. What are Spot Instances and On-Demand Instances?
26. Explain Connection Draining.
27. What is a Stateful and a Stateless Firewall?
28. What is Power User Access in AWS?
29. What is an Instance Store Volume and an EBS Volume?
30. What are Recovery Time Objective (RTO) and Recovery Point Objective (RPO) in AWS?
31. Is there a way to upload a file that is greater than 100 MB in Amazon S3?
32. Can you change the Private IP Address of an EC2 instance while it is running or in a stopped state?
33. What is the use of lifecycle hooks in Auto Scaling?

34. What are the policies that you can set for your users' passwords?

Prepare basic design concepts

- Modularity
- Separation of concerns
- Reusability
- Scalability
- Maintainability
- Coupling
- Cohesion
- SOLID basics
- Design patterns basics
 - Singleton
 - Factory
 - Observer